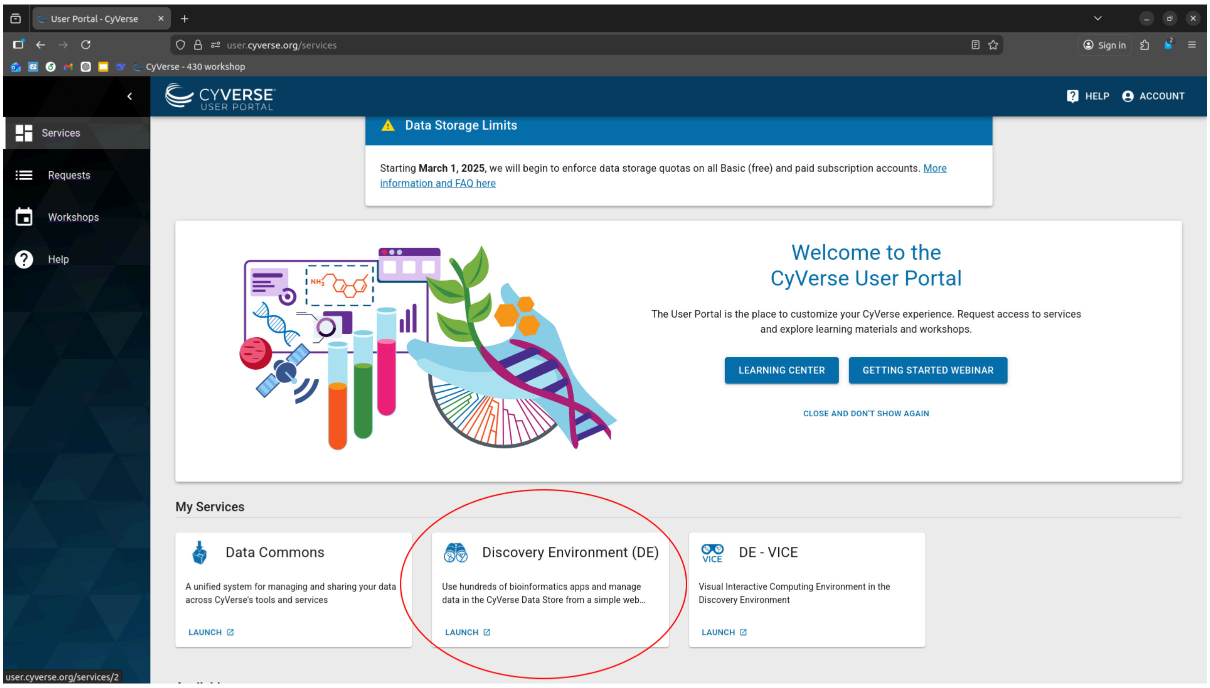
Tutorial: Running Mutation & PTM Notebooks on CyVerse

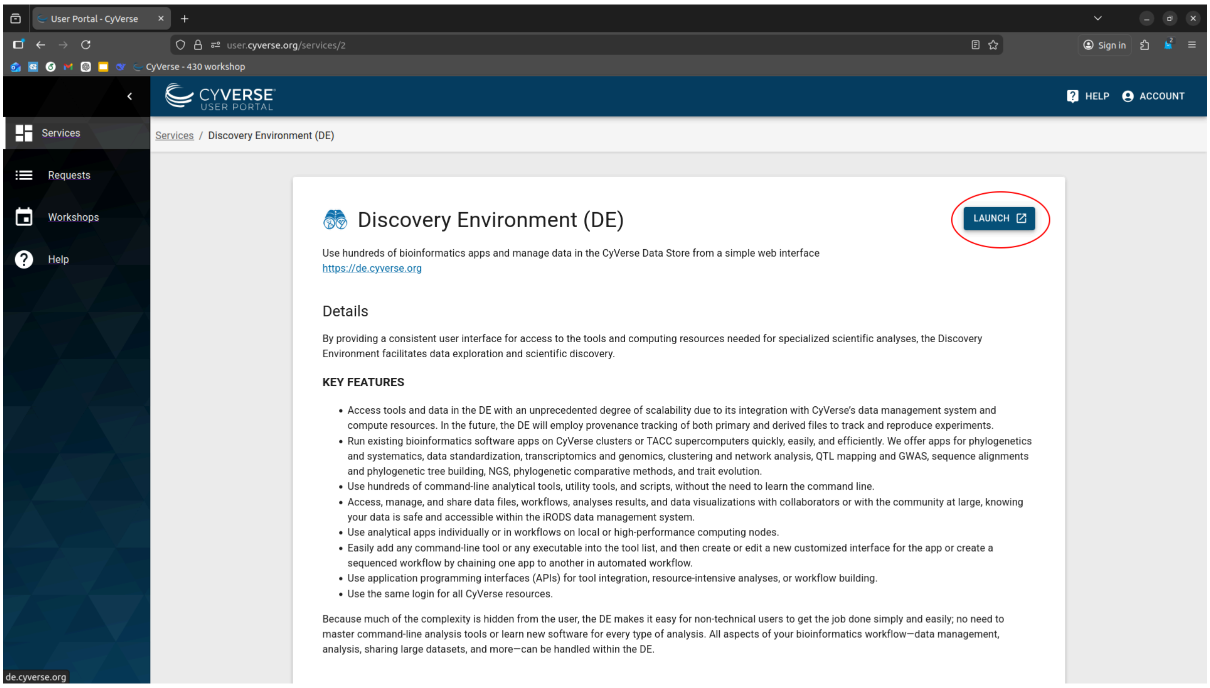
This guide will walk you through launching the Discovery Environment and configuring the specialized Python kernel required for your protein analysis.

Step 1: Launch the Discovery Environment

1. Log in to the **CyVerse User Portal**.
2. On the **Services** page, locate the **Discovery Environment (DE)**.



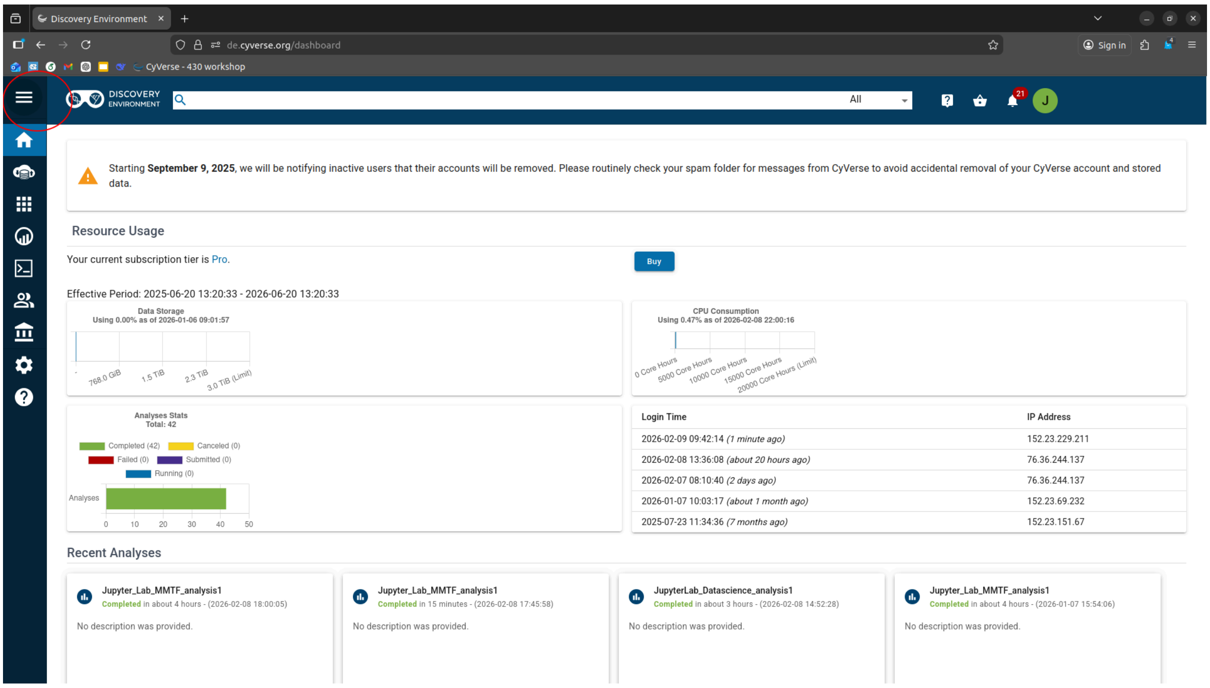
3. Click the blue **LAUNCH** button.



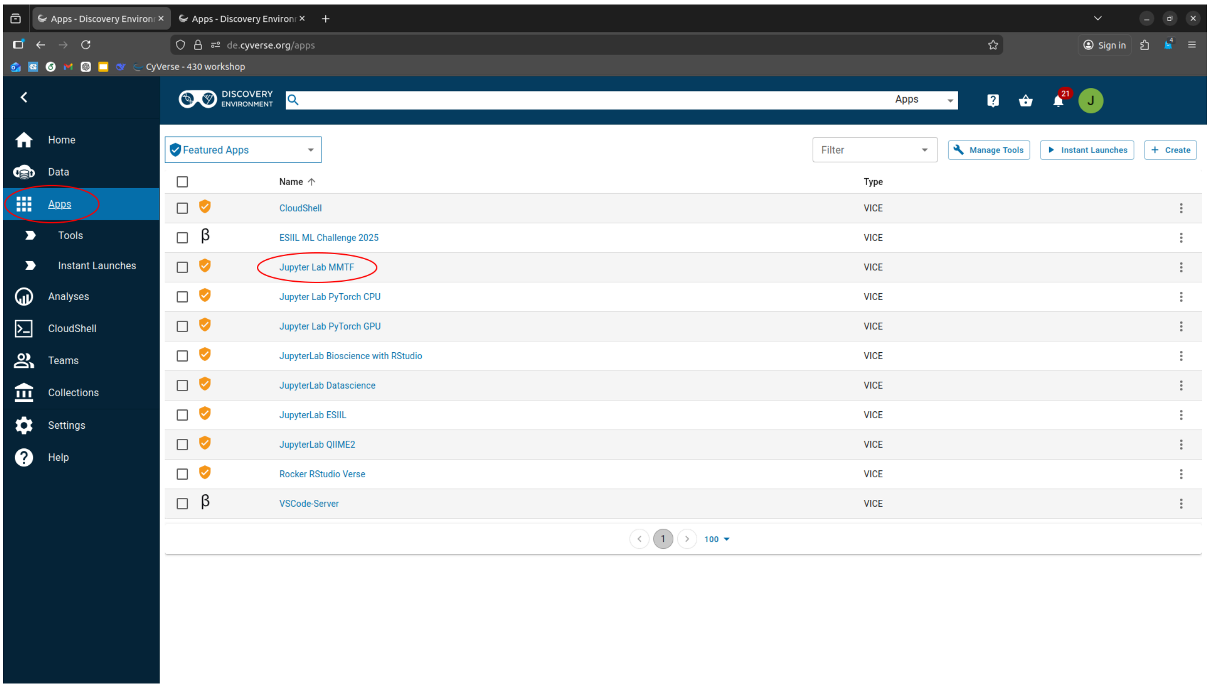
Step 2: Open and Launch Jupyter Lab MMTF

Once the Discovery Environment opens in a new tab:

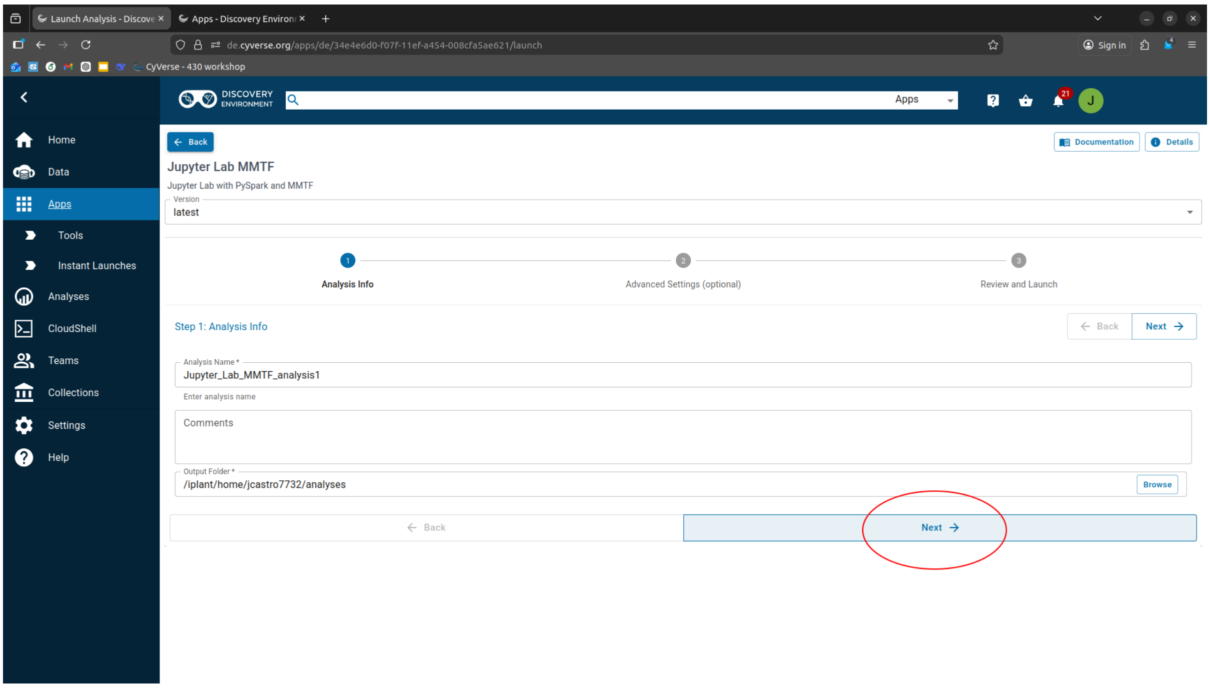
1. **Open Apps:** Click the **three horizontal lines** (sidebar menu) on the far left and select **Apps**.



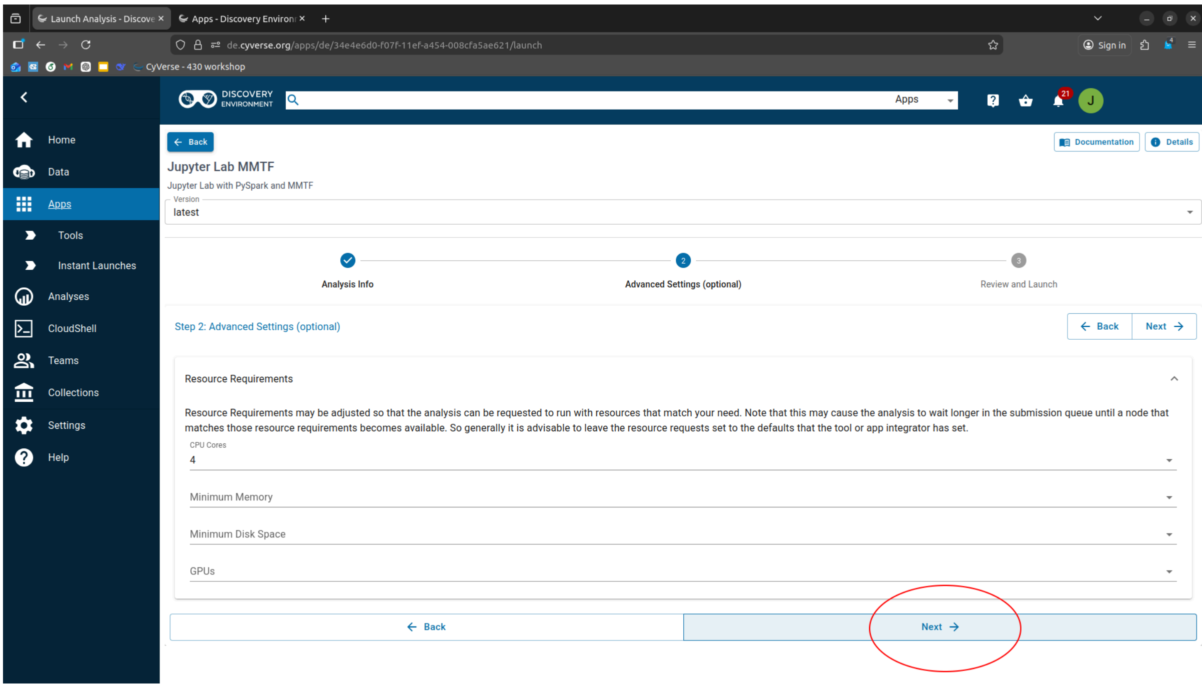
2. **Find the App:** In the search bar or list, click on **Jupyter Lab MMTF**.



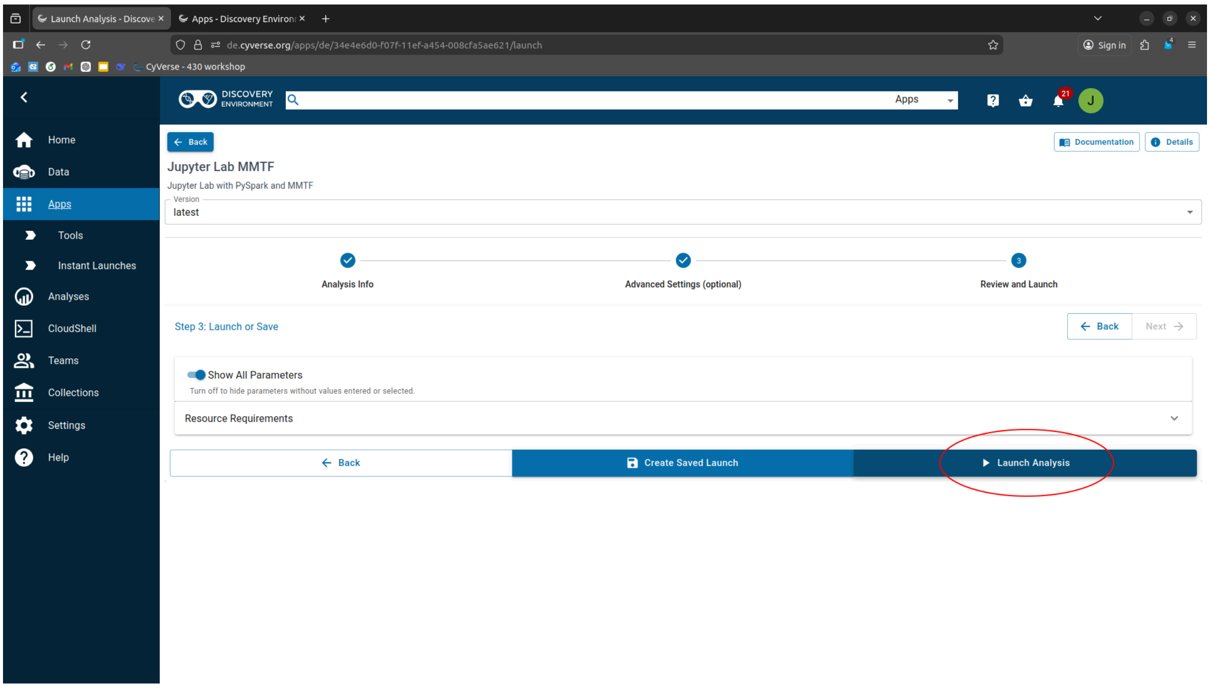
3. **Navigate Setup:** * On the "Analysis Info" screen, click **Next**.



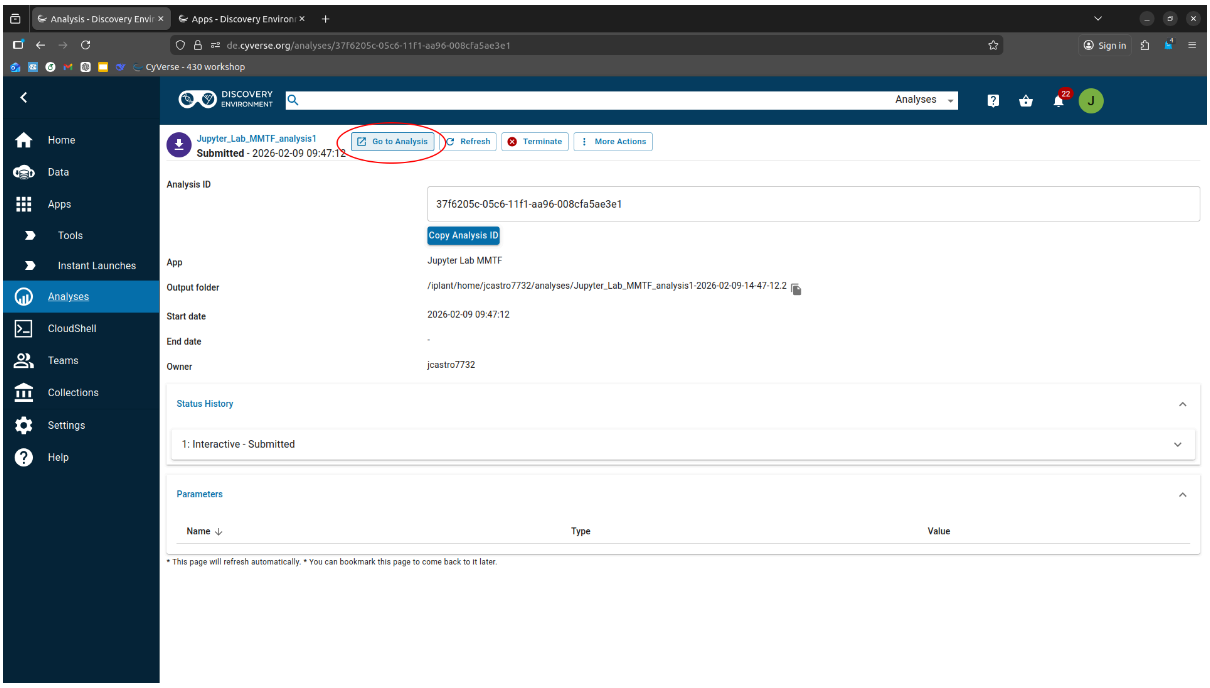
- On the "Advanced Settings" screen, click **Next**.



4. **Launch:** Click **Launch Analysis** at the bottom right.



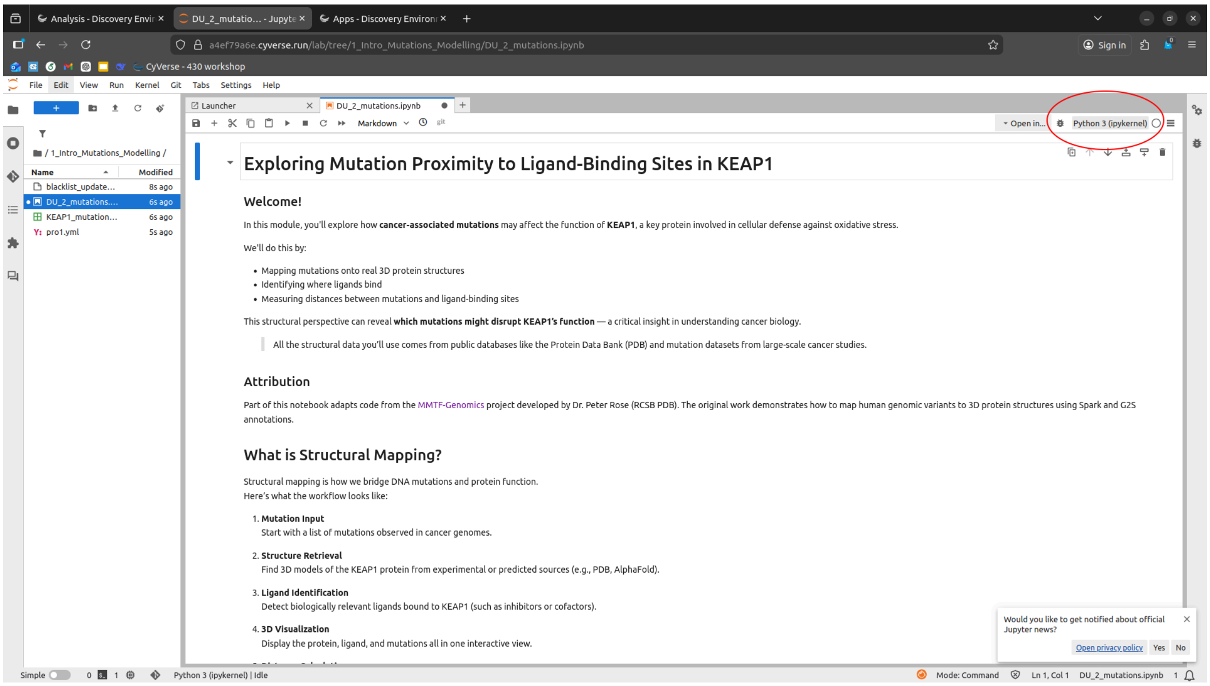
5. **Access Session:** A status window will appear. Click **Go to Analysis** to enter your Jupyter workspace.



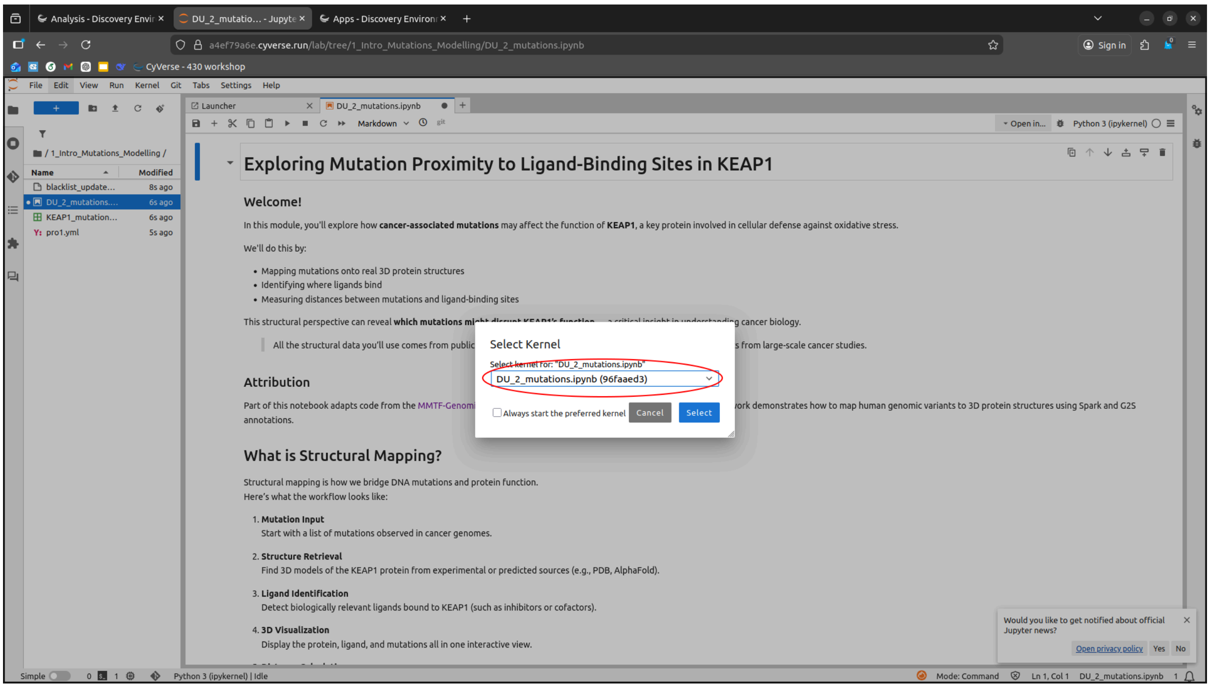
Step 3: Configure the MMTF-PySpark Kernel

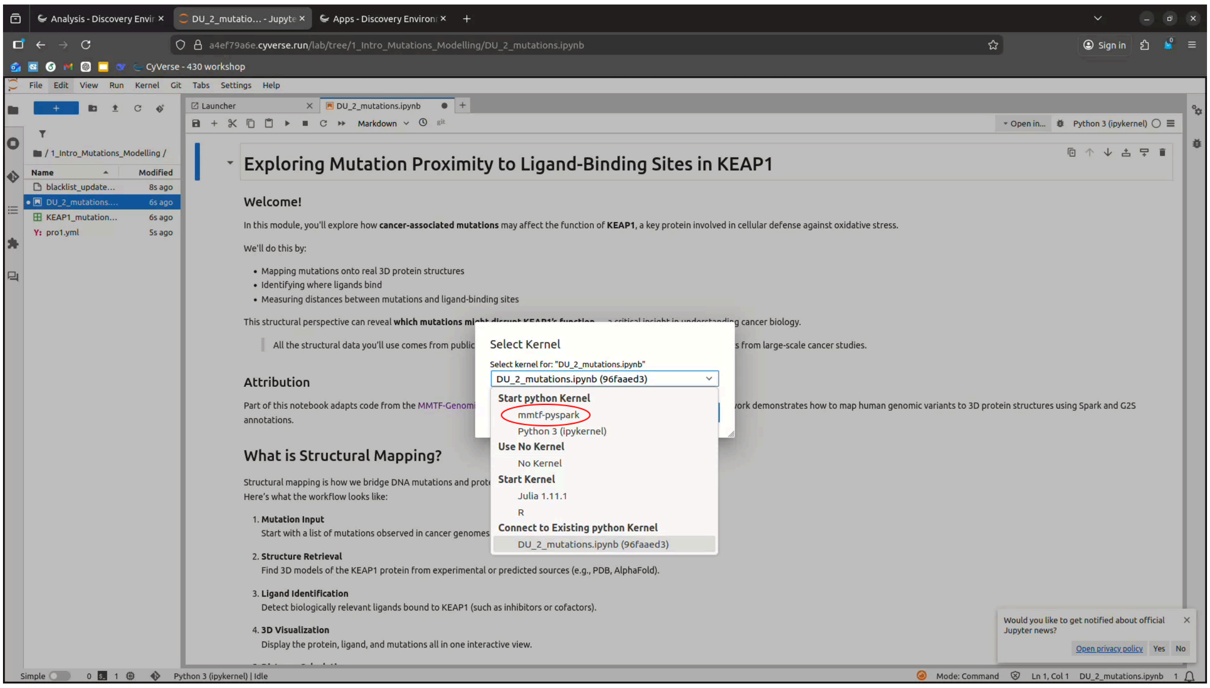
After you have uploaded your notebooks into the Jupyter interface, you must switch to the correct computing environment (the "Kernel").

1. **Open Kernel Settings:** In the top-right corner of the notebook interface, click the text that says **Python 3 (ipykernel)**.



2. **Select Proper Kernel:** In the pop-up "Select Kernel" menu, click the dropdown and choose **mmtf-pyspark**.





3. **Confirm:** Click the blue **Select** button.
4. **Verify:** Ensure the top-right corner now displays **mmtf-pyspark**. You are now ready to run your code cells.

Summary Checklist

- [ ] Launched Discovery Environment?
- [ ] Selected **Jupyter Lab MMTF** (not standard Jupyter)?
- [ ] Clicked **Go to Analysis**?
- [ ] Switched kernel to **mmtf-pyspark**?